

Question number	Answer	Additional guidance	Mark
7(a)(i)	Choose an item. • B is the correct answer A is not the correct answer as two statements are correct C is not the correct answer as energy in muscle contraction is not used to release ADP and Pi from ATP D is not the correct answer as two statements are correct		(1)

Question number	Answer	Additional guidance	Mark
7(a)(ii)	A description that includes four of the following points: - calcium ion channels open in the sarcoplasmic reticulum / calcium ions released {from sarcoplasmic reticulum / into sarcoplasm} (1) - allowing myosin to bind to actin / (actin-myosin) cross bridge formed (1) - ADP and Pi are released resulting in a power stroke / ADP and Pi are released resulting in actin pulled {over myosin / towards the M line/ towards the centre} (1) - ATP binds (to myosin head) and {crossbridge is broken / myosin head detaches} (1) - ATP is hydrolysed (1) (providing energy) to change the myosin (head) {shape / position} (1)	mps need to be in logical order to gain credit accept calcium ions bind to troponin / calcium ions cause (myosin) binding sites to be exposed accept ATP broken down into ADP and Pi accept myosin (head) returns to its original position do not accept the power stroke	(4)

Question number	Answer	Additional guidance	Mark
7(b)(i)	An answer that includes two of the following points: • {muscle T is the largest muscle / muscle F is the smallest muscle} (in both species) (1) • (mean) muscle mass is larger in the Zimbabwean blue / the Zimbabwean blue muscles are larger (1) • correct conclusion regarding SD and mean mass of muscles (1)	accept order of muscles in size $T > G > F$ accept converse accept for all muscles or a stated muscle e.g. • no significant difference (between the two ostriches) for muscle T as SD overlap • significant difference (between the two ostriches) for muscle {F / G} as SD do not overlap • no significant difference between muscle G and F as SD overlap	(2)

Question number	Answer	Additional guidance	Mark
7(b)(ii)	An answer that includes two of the following points: low validity (of conclusions) because: - muscle T in both types of ostrich as the SD overlap (1) {very few / small sample size of} Zimbabwean blue (1) - stated variables may not have been controlled (1) - it was only one study (1) conclusions valid because: - muscle {F / G} for SAB and Zimbabwean blue has {significant difference / no SD overlap} (1)	ignore no repeats accept the numbers of Zimbabwean Blue and South African Black are not the same allow results may be anomalous e.g. nutrition / physical activity / age / gender accept small {SD/error bars} indicating {repeatability / validity / less variability in data}	(2)

Question number	Answer	Additional guidance	Mark
7(b)(iii)	A calculation showing the following steps: • calculation of difference between muscle mean mass (1) • calculation of percentage difference and answer to two decimal places (1)	0.59 – 0.74 or -0.15 or (0.74 – 0.59) = 0.15 (-0.15 ÷ 0.665) x 100 = (-)22.56 Correct answer scores full marks ecf for second marking point if uses added or subtracted SDs ecf if use wrong muscle but calculation correct and to two decimal places = 1 mark eg. for muscle T = 24.20 Muscle F = 22.22	(2)

These are the most common responses but please always use the mark scheme

1 mark	2 marks
0.59 – 0.74 (=0.15)	(-)22.56
0.74 – 0.59 (= -0.15)	
(-) 0.15	
(-) 22.6	
(-)13.89	
(-)5.80	
(-)40.63	
(-)32.79	
(-)24.20 (muscle T)	
(-)22.22 (muscle F)	

Question number	Answer	Additional guidance	Mark
7(c)	An explanation that includes three of the following points: - thermoreceptors detect changes in temperature (1) (impulses sent to) thermoregulatory control centre / heat {gain / loss} centre (1) - hypothalamus sends impulses {via motor neurones / via sympathetic neurones / to appropriate named effector} (1) - correct response (by effectors) to {increase heat energy gain / reduce heat energy loss} (1)	reject signal accept hypothalamus has thermoreceptors / temperature sensor / temperature receptor accept hypothalamus receives impulses from {thermoreceptors / temperature sensors / temperature receptor} ignore thermoregulators ignore control centre unqualified e.g. skeletal muscles, sweat glands, hair erector muscles, liver, smooth muscles in skin blood vessels e.g. vasodilation, metabolism of brown fat, (increased / decreased) sweat production	(3)

